SAT Hard Question 1

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Question: Greatest Possible Value

In the given expression, a and c are positive constants. If $px + q$ is a factor of the expression:

$$ax^2 + 176x + c = (px + q)(rx + s),$$

where p and q are positive constants, what is the greatest possible value of ac?

Answer Explanation: Maximum Value of ac

Correct Answer: 7744

Solution:

- We use the discriminant condition:

$$\Delta = b^2 - 4ac > 0.$$

- Substituting $b = 176$:

$$176^2 - 4ac > 0.$$

- Simplifying:

$$4ac \leq 176^2.$$

- Solving for ac:

$$ac \leq 7744.$$

Final Answer: $\boxed{7744}$.

Question: Finding the Greatest Possible Value of b

In the given expression:

$$34z^{18} + bz^9 + 30,$$

b is a positive integer.

If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b?

Answer Explanation: Maximum Value of b

Correct Answer: 1021

Solution:

- Compute b using factor constraints:

$$b = 34 \times 30 + 1.$$

- Performing the calculation:

$$b = 1021.$$

Final Answer: $\boxed{1021}$.

Question: Finding the Value of $a + c$

The expression:

$$6x^4 + 17x^2 + 7$$

can be rewritten as:

$$(3x^2 + a)(2x^2 + b),$$

where a and b are positive integers, or as:

$$(3x^2 + c)(2x^2 + d),$$

where c and d are positive non-integers.

What is the value of $a + c$?

Answer

We are given:

$$6x^4 + 17x^2 + 7$$

We want to factor this as:

$$(3x^2 + a)(2x^2 + b)$$

Multiply:

$$(3x^2 + a)(2x^2 + b) = 6x^4 + (2a + 3b)x^2 + ab$$

Match coefficients:

$$6x^4 + (2a + 3b)x^2 + ab = 6x^4 + 17x^2 + 7 \Rightarrow \begin{cases} ab = 7 \\ 2a + 3b = 17 \end{cases}$$

Final Answer: $\boxed{\dfrac{17}{2}}$

Question: Finding the Smallest Possible Value of ab

The given quadratic function:

$$54x^4 + 219x^2 + 105$$

has factors in the form:

$$(k)(ax^2 + b)(cx^2 + d).$$

If a, b, c, d, and k are integers, what is the smallest possible value of ab?

Answer

Final Answer: $\boxed{14}$

Question: Finding $a - b - c$

The equation of a parabola is written in the form:

$$y = ax^2 + bx + c,$$

where a, b, and c are constants.

When graphed in the xy-plane, the parabola has vertex $(-1, 4)$ and does not intersect the x-axis.

Which of the following could be the value of $a - b - c$? **Options:**

A 2

B 4

C -1

D -6

Answer Explanation: Finding $a - b - c$

Correct Answer: D

Solution:

- Since the parabola does not intersect the x-axis, the discriminant condition is:

$$\Delta = b^2 - 4ac < 0.$$

- Solving the given system:

$$\boxed{D}.$$

Question: Maximum Value of b

A quadratic function with vertex $(1, 32)$ can be written as:

$$-ax^2 + bx + c \quad (a, b, c \text{ are all positive integers}).$$

What is the maximum value of b?

Answer Explanation: Maximum Value of b

Correct Answer: 62

Solution:

- Given:

$$b = 2a \quad \text{and maximum } a + c = 32.$$

- Solving for $b_{\max}$:

$$b_{\max} = \boxed{62}.$$

Question: Converting Acceleration Units

An object's speed is increasing at a rate of $7.7\,\mathrm{m/s^2}$.

What is this rate in miles per minute squared, rounded to the nearest tenth?
(Use $1\,\text{mile} = 1,609\,\text{meters.}$)

Options:

1. 0.3
2. 17.2
3. 206.5
4. 209

Answer Explanation: Converting Acceleration Units

Correct Answer: 17.2

Solution:

- Convert units:

$$7.7\,\frac{\text{m}}{\text{s}^2} \times \left(\frac{60\,\text{s}}{1\,\text{min}}\right)^2 \times \frac{1\,\text{mile}}{1609\,\text{m}}.$$

- Simplify:

$$7.7 \times \frac{3600}{1609}.$$

- Approximate:

$$\approx 17.2\,\frac{\text{miles}}{\text{min}^2}.$$

Final Answer: $\boxed{17.2}$.

Question: Finding $f(5)$

The function f is defined by:

$$f(x) = ab^{x/n},$$

where $a, b,$ and n are constants, and b, n are integers.

If $f(2) = 6$ and $f(4) = 150$, what is the value of $f(5)$?

Answer Explanation: Finding $f(5)$

Correct Answer: 750

Solution:

- Given:

$$f(2) = 6 \implies a \cdot b^{2/n} = 6.$$

$$f(4) = 150 \implies a \cdot b^{4/n} = 150.$$

- Divide equations:

$$\frac{b^{4/n}}{b^{2/n}} = b^{2/n}, \quad b^{2/n} = 25 \implies b^{1/n} = 5.$$

- Solve for $f(5)$:

$$f(5) = a \cdot b^{5/n} = \frac{6}{25} \cdot (5^5) = \frac{6}{25} \times 3125.$$

Final Answer: $\boxed{750}$.

Question: Positive Difference Between a and b

The quadratic equation graph of $y = f(x)$ in the xy-plane has:

- A vertex at $(4, -7)$.
- Contains the point $(3, -9)$.
- Has a y-intercept at $(0, a)$.
- The graph of $y = 6 \cdot f(x)$ has a y-intercept at $(0, b)$.

What is the positive difference between a and b?

Answer Explanation

Correct Answer: 195

Solution:

- Equation of the parabola:

$$f(x) = k(x-4)^2 - 7.$$

- Substitute $(3, -9)$ to find k:

$$-9 = k(3-4)^2 - 7 \implies k = -2.$$

- Thus, the equation becomes:

$$f(x) = -2(x-4)^2 - 7.$$

- Find a:

$$a = f(0) = -39.$$

- Find b for $y = 6 \cdot f(x)$:

$$b = 6 \cdot (-39) = -234.$$

Answer Explanation

Correct Answer: 195

Solution:

- Compute the positive difference:

$$|a - b| = |234 - 39| = \boxed{195}.$$

Question: Finding $a + b$

The function f is defined by:

$$f(x) = ax^2 + bx + c,$$

where a, b, and c are constants.

The graph of $y = f(x)$ in the xy-plane passes through the points $(9, 0)$ and $(-2, 0)$.

If a is an integer greater than 1, which of the following could be the value of $a + b$?

Options:

A 8

B 7

C -6

D -12

Answer Explanation: Finding $a + b$

Correct Answer: -12

Solution:

- Factorize the quadratic equation:

$$f(x) = a(x-9)(x+2) = a(x^2 - 7x - 18).$$

- From this equation:

$$b = -7a, \quad c = -18a.$$

- For $a \geq 2$, calculate:

$$a + b = a - 7a = -6a.$$

- Possible values:

$$a + b = -12 \quad \text{when } a = 2.$$

Final Answer: $\boxed{-12}$.

Question: Finding a Factor of $x + 2b$

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

$$f(x) = \begin{cases} A : 3x^2 + 7x + 14b, \\ B : 3x^2 + 22x + 14b, \\ C : 3x^2 + 30x + 14b, \\ D : 3x^2 + 37x + 14b. \end{cases}$$

Answer Explanation: Finding a Factor of $x + 2b$

Correct Answer: D

Solution:

- By the factor theorem:

$$\text{If } (x - c) \text{ is a factor of } f(x), \text{ then } f(c) = 0.$$

- Let $x + 2b$ be a factor of $f(x)$. Substitute $x = -2b$:

$$f(-2b) = 0.$$

- Calculate for each option, and we find that when $b = 5$, only:

$$f(x) = 3x^2 + 37x + 14b$$

  satisfies $f(-2b) = 0$.

Final Answer: $\boxed{D}$.